

Model Comparison Report

This report evaluates the performance of Random Forest and XGBoost classifiers in predicting bank deposit subscriptions, following rigorous hyperparameter tuning and threshold optimization. The Random Forest model, optimized via GridSearchCV, established a strong baseline with an AUC of 0.7991. At a standard 0.5 threshold, it prioritized precision (0.66) but suffered from a very low recall (0.23), failing to identify the vast majority of potential subscribers. However, by adjusting the threshold to 0.3, it achieved a more functional balance for the business, reaching a 0.49 recall and 0.47 precision.

The XGBoost model was subjected to a granular nested loop search, yielding a superior AUC of 0.8024. This model consistently outperformed Random Forest across various sensitivity levels. Specifically, at the 0.3 threshold, XGBoost delivered a precision of 0.49 and a recall of 0.48. This indicates that XGBoost is more efficient at distinguishing true potential subscribers from the "noise," ensuring that nearly half of the targeted calls result in a successful conversion—a critical metric for maintaining call center morale and operational efficiency.

The cross-validation analysis for XGBoost further confirms its reliability for a production environment. With an Average AUC of 0.7862 and a very low Standard Deviation of 0.0064, the model demonstrates remarkable stability across different data folds. This low variance proves that the model is not overfitted to specific patterns in the training set but is likely to maintain its predictive power when applied to new, unseen customers. The consistency of these "scores per fold" provides the necessary confidence for a real-world rollout.

Feature Importance analysis revealed that the model is heavily driven by macroeconomic and historical engagement indicators. The variable `nr_employed` (number of employees) emerged as the dominant predictor with a weight of 0.48, followed by `emp.var.rate`, `pdays`, and `euribor3m`. This suggests that the customer's likelihood to invest is deeply tied to broader economic stability and the timing since their last contact. By leveraging these features, the XGBoost model transforms raw data into a strategic map for the marketing department, identifying the exact economic "window" where a customer is most likely to commit.

Recommendation: XGBoost is the superior model for production. It offers a slightly higher AUC and better precision at lower thresholds than Random Forest. We recommend deploying XGBoost with a 0.3 decision threshold. This configuration provides the optimal "sweet spot": it captures nearly 50% of all potential subscriptions while ensuring that the sales team only contacts leads with a high probability of success. This strategy maximizes total deposit volume while minimizing the operational costs associated with unsuccessful marketing attempts.